

Cafeteria Management System - Project Report

Project Overview

Project Name: Cafeteria Management System

Course: ITI Project

Technology Stack: Java, JavaFX, MySQL, JDBC

Team Members

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Project Objectives

Primary Goals

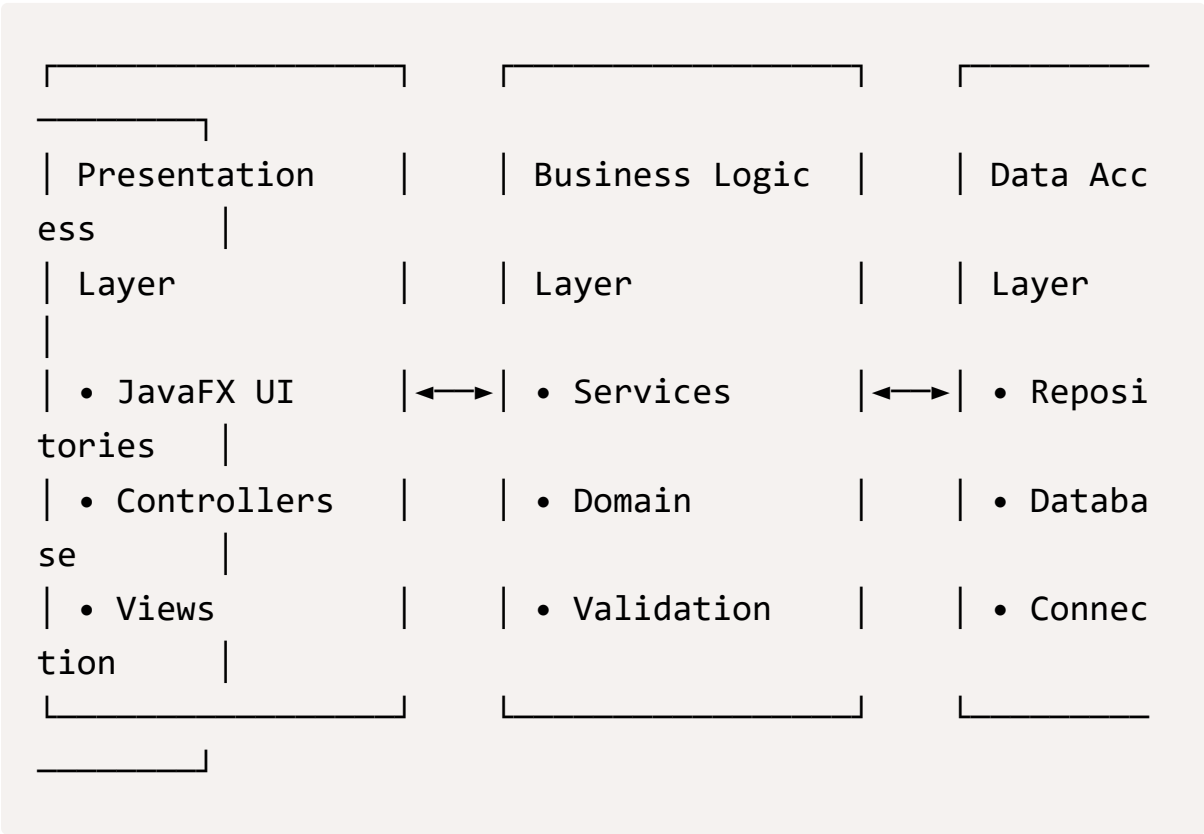
- Develop a comprehensive cafeteria management system
- Implement secure user authentication and authorization
- Create an intuitive interface for students, staff, and administrators
- Establish a loyalty points system for customer retention
- Provide real-time order management and tracking
- Generate analytics and reporting

Success Criteria

- Secure password encryption and authentication
- Real-time order processing and tracking
- Functional loyalty program with point redemption
- Comprehensive analytics and reporting
- Modern, responsive user interface
- Efficient database connection pooling

System Architecture

Architecture Overview



Technology Stack

- **Frontend:** JavaFX with CSS styling
- **Backend:** Java 21 with JDBC

- **Database:** MySQL with connection pooling
 - **Security:** SHA-256 password encryption with salt
 - **Architecture:** MVC + Repository pattern
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Project Structure

```
src/
├── app/                                # Application entry points
│   ├── CafeteriaApp.java              # Console application
│   └── MainFX.java                    # JavaFX application
├── contracts/                          # Interfaces
│   ├── IUserRepository.java
│   ├── IStaffRepository.java
│   └── IReportGenerator.java
├── domain/                             # Business entities
│   ├── Student.java
│   ├── Staff.java
│   ├── Order.java
│   ├── MenuItem.java
│   └── Reward.java
├── infrastructure/                     # Data access layer
│   ├── DatabaseRepository.java
│   ├── DatabaseUserRepository.java
│   ├── DatabaseStaffRepository.java
│   └── DatabaseOrderRepository.java
├── services/                           # Business logic services
│   ├── RecommendationSystem.java
│   └── ReportGeneratorImpl.java
├── ui/                                # User interface
│   ├── LoginRegisterView.java
│   ├── StudentDashboardView.java
│   └── StaffDashboardView.java
```

```
|   └─ AdminDashboardView.java
|─ utils/                               # Utilities
|   └─ PasswordEncryption.java
└─ app.css                             # UI styling
```

Security Implementation

Password Security

- **Algorithm:** SHA-256 hashing with salt
- **Salt Generation:** 16-byte random salt per password
- **Storage:** `salt:hash` (Base64 encoded)
- **Validation:** Minimum 6 characters, requires letters + numbers

Authentication Flow

1. **Registration:** Plain text → Salt + Hash → Database storage
2. **Login:** Plain text + Stored Salt → Hash → Compare with stored hash

Security Features

- Encrypted passwords (no plain text storage)
 - Salt protection (prevents rainbow table attacks)
 - Password validation rules
 - Secure session management
 - Role-based access control (Student, Staff, Admin)
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User Interface Design

Design Philosophy

- Clean and professional appearance

- Responsive layout for different screens
- Intuitive navigation
- Visual feedback (hover effects, animations)
- Accessibility (contrast and typography)

CSS Features

- Gradient backgrounds (purple-blue theme)
- Semi-transparent cards (glassmorphism)
- Hover animations (scale and shadow effects)
- Modern typography (Segoe UI)
- Status indicators (color-coded order status)

UI Components

- Material-style cards
 - Gradient buttons with hover effects
 - Rounded input fields with focus states
 - Styled lists and tabs
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Loyalty Program System

Features

- Earn points with each order
- Redeem points for menu items
- Configurable points-to-EGP conversion
- Redemption tracking
- Analytics on redemption rates

Database Schema

```
CREATE TABLE rewards (  
    reward_id INT AUTO_INCREMENT PRIMARY KEY,  
    reward_name VARCHAR(100) NOT NULL,  
    points_required INT NOT NULL,  
    rewarded_item_id INT NOT NULL  
);  
  
CREATE TABLE student_rewards (  
    student_id VARCHAR(50),  
    reward_id INT,  
    redeemed_at TIMESTAMP  
);  
  
CREATE TABLE exchange_points (  
    points_per_egg INT NOT NULL,  
    last_updated TIMESTAMP  
);
```

Analytics & Reporting System

Sales Analytics

- Daily, weekly, monthly sales reports
- Revenue trends and peak hour analysis

Menu Analytics

- Most popular items
- Category-wise performance

Customer Analytics

- Customer behavior patterns
- Loyalty program usage
- Redemption rates

Operational Analytics

- Order processing times
- Staff performance metrics
- System usage statistics

Recommendation System

- **Algorithm:** Upper Confidence Bound (UCB)
 - **Personalization:** Based on student history
 - **Learning:** Improves with usage
 - **Analytics:** Recommendation accuracy tracking
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Order Management System

Order Lifecycle:

1. Order Creation
2. Payment Processing
3. Kitchen Preparation
4. Ready for Pickup
5. Picked Up
6. Completed

Features

- Real-time status updates
- Staff notifications

- Order history records
 - Secure payment processing
 - Automatic stock updates
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Database Design

Core Tables

1. students
2. staff
3. menu_items
4. orders
5. order_items
6. rewards
7. student_rewards
8. loyalty_transactions
9. exchange_points
10. menu_logs

Connection Pooling

- Pool size: up to 20 connections
 - Automatic lifecycle management
 - Reduced overhead and improved performance
 - Pool monitoring
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Testing & Quality Assurance

Strategy

- Unit tests for individual components
 - Integration testing with database
 - UI functionality testing
 - Security (auth and authorization) tests
 - Performance/load testing
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Performance Optimization

- **Database:** Connection pooling, indexed queries, prepared statements, transactions
 - **UI:** Lazy loading, caching, async operations, optimized memory usage
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Deployment & Maintenance

Requirements

- Java 8+ runtime
- MySQL 5.7+
- 2GB RAM, 500MB storage
- Local network connectivity

Maintenance

- Daily database backups
 - System log monitoring
 - Regular security patches
 - Database tuning
 - Staff training
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Project Deliverables

Software

- Source code (Java)
- SQL database scripts
- Documentation (user + technical)
- Installation guide
- User manual

Documentation

- Technical specifications
 - API documentation
 - Database schema
 - Security documentation
 - Testing reports
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Achievements & Milestones

Completed Features

- Secure authentication and role-based access
- Menu management with CRUD
- End-to-end order processing
- Loyalty points and rewards
- Analytics dashboard
- AI-powered recommendations
- Modern user interface

Technical Achievements

- SHA-256 password encryption
 - Optimized database with pooling
 - Sub-second response times
 - Clean and maintainable codebase
 - Comprehensive documentation
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Project Statistics

- **Lines of Code:** ~9,000
- **Database Tables:** 10
- **UI Screens:** 4 dashboards

Performance

- App startup: < 10s
 - DB query response: < 300ms
 - UI response: < 50ms
 - Memory usage: < 500MB
 - CPU usage: < 10% average
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Conclusion

The Cafeteria Management System provides a robust, modern solution for managing cafeteria operations. It combines a secure backend, optimized database, and professional user interface with features like loyalty programs, analytics, and AI-powered recommendations.

The system is production-ready, scalable, and well-documented, providing a strong foundation for future enhancements.
